Atria: Personalized, Safe Transportation

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User Research

Interview

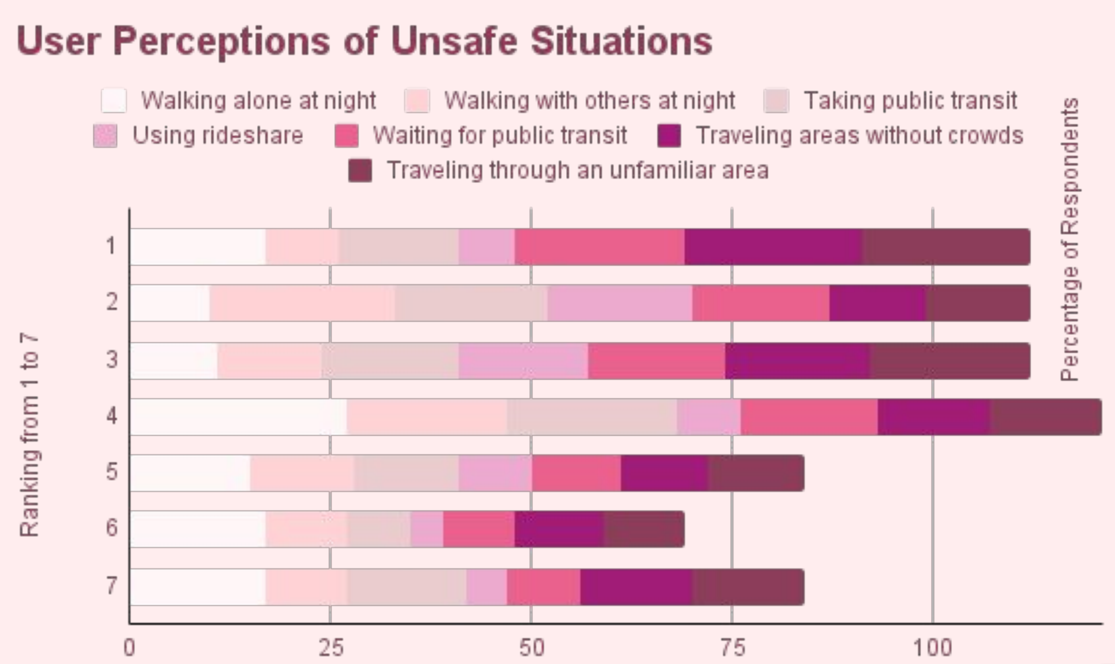
32 students and new grad ages 16-25

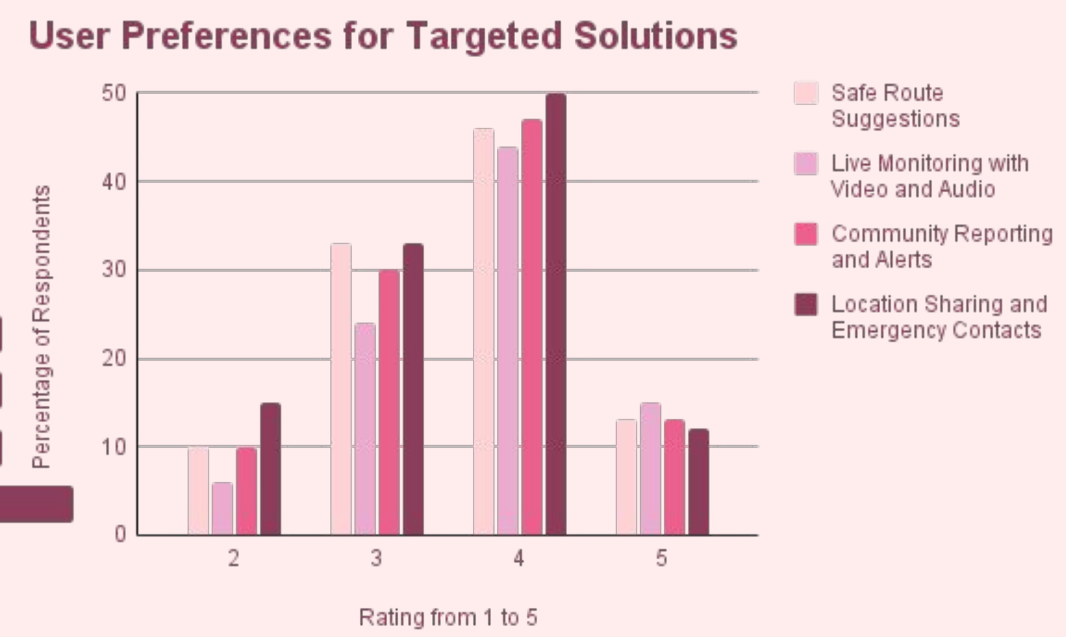
Survey

25 students and new grad ages 16-25

Observational Study

1 hour study of UCLA students and LA residents taking **public transportation** around West Hollywood



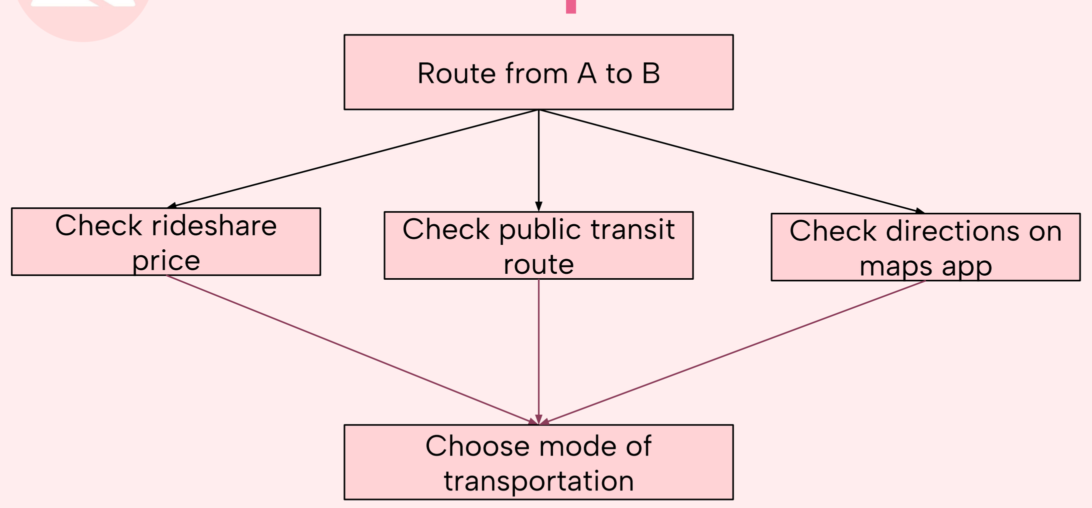


Analysis

Personas

Women and people of color ages 16-25 who ever (frequently or rarely) travel by car, rideshare, or public transit. Different people prioritize different travel criteria, such as lighting, traffic, crime, pedestrians, etc., to different degrees.

Process Map



Challenges

1. **Apps**: Users currently open many different apps to compare their routing options
2. **Decision-Making:** There is no concrete way in which a user decides which mode of transportation and route to take
   a. **Preferences:** Some travel criteria users care about, like lighting and crime, are not incorporated in routing options
   b. **Personalization:** Everybody is recommended the same routes regardless of their personal preferences above

Problem Statement

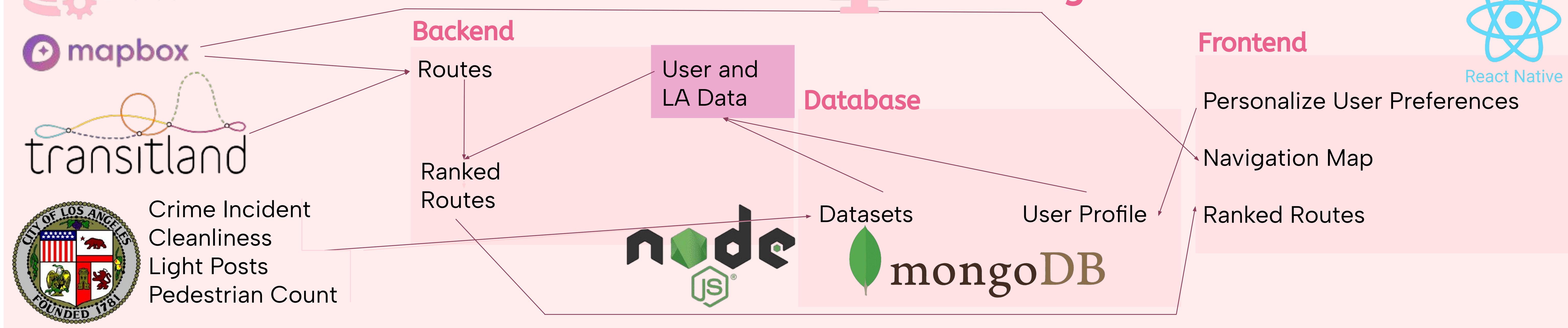
Women and people of color in particular are concerned about **safety** when traveling. However, current transportation apps mainly prioritize speed and efficiency when navigating. Users need a method for **comparing alternative routes** depending on their **personal preferences** towards lighting, speed, cost, cleanliness, crime, etc.

System Overview

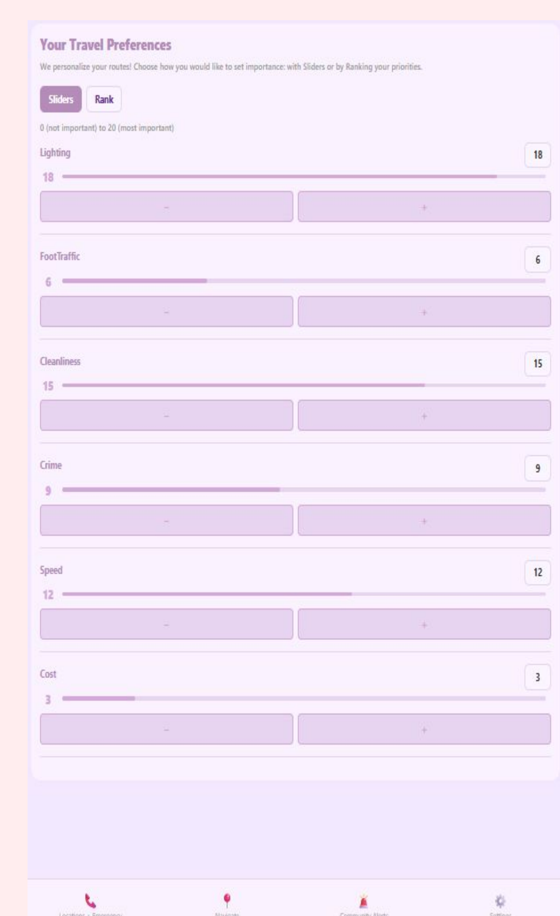
Atria is an all-in-one Los Angeles transportation app that ranks **all possible routes** based on each user's personal safety preferences, including lighting, foot traffic, crime history, and cleanliness. Instead of choosing routes solely by speed, Atria empowers users to **customize** what matters most to them and receive **tailored, safety-forward navigation**.

Data

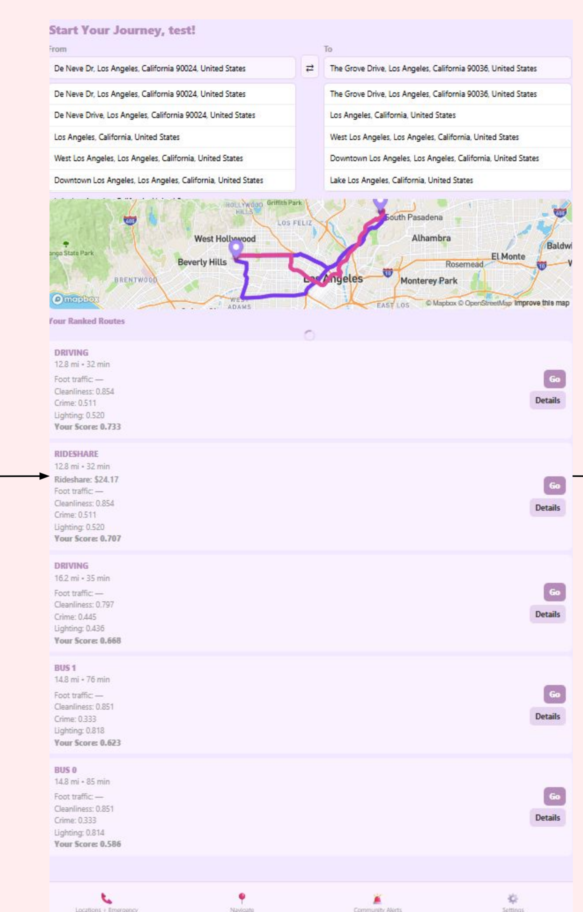
Technologies



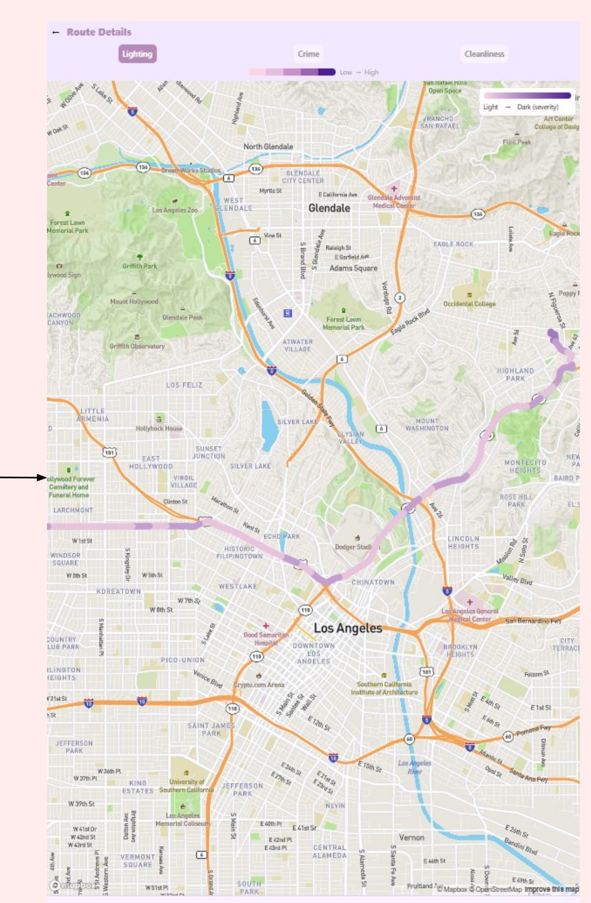
User Interface



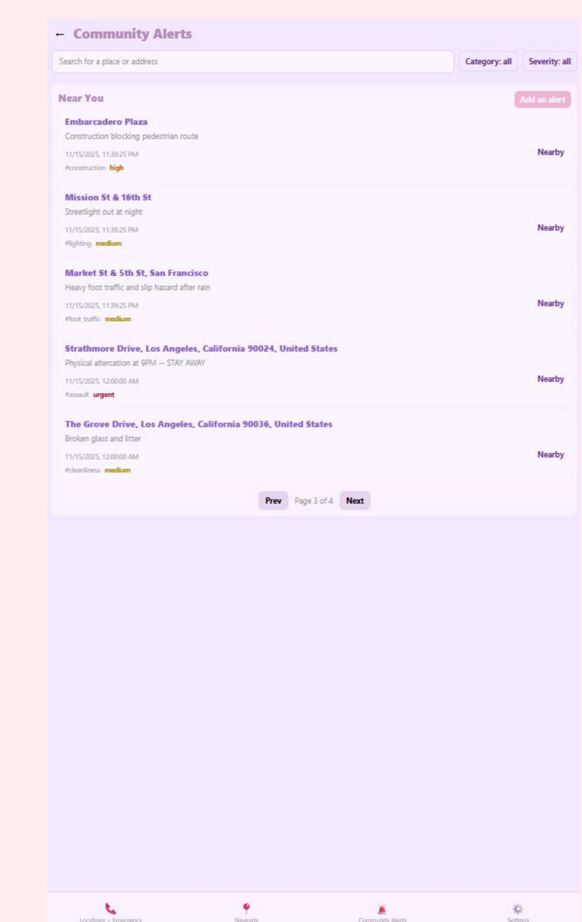
Personalize Travel Preferences



Ranked Navigation Routes by Travel Preferences



Further Details per Route per Travel Criteria



Community Alerts

Evaluation

- Easy to **personalize** and see route re-rankings in real time
- Meets user demand for **safety criteria** in transportation apps
- **Transparent scoring** and information on safety criteria
- Real-time **community information** and forum-like alerts

Next Steps

- **Caching** and faster **latency** for improved UX
- More **real-time** safety datasets
- **Travel buddy** feature for increased safety